SESSION 3

Data Preparation and Feature Engineering for Applied AI

Missing data • Outliers • Scaling • Encoding • Leakage • Domain features

Applied AI Design Workshop for Faculty
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Goal

turn raw disciplinary data into a trustworthy modeling table.

Session outputs

- Data-readiness checklist
- Preprocessing plan
- Leakage-risk decision
- Candidate feature list
- Minimal coding pattern

Core idea

Model quality is bounded by data quality and representation. Better features often beat bigger models.

Activity roadmap

Audit

Rows, types, missingness, target

Clean

Missing values, outliers, categories

Transform

Scaling and encoding

Protect

Leakage and split strategy

Engineer

Domain and time-series features

Document

Preprocessing plan

Facilitator note

Keep every preprocessing decision grounded in the prediction point: what would be known before the model is asked to predict?

Data preparation is the bridge between problem framing and model training.

Why data preparation matters

Applied AI fails when the data table does not match the real decision context.

Raw data

- Different units
- Missing values
- Text labels
- Sensor noise
- Time stamps
- Batch identifiers

Modeling table

- One row per observation
- Clean inputs
- Target separated
- Encoded categories
- Scaled numeric fields
- No leakage

Defensible model

- Baseline comparison
- Valid split
- Interpretable metrics
- Reproducible decisions
- Clear limitations

Preprocessing is not clerical cleanup — it is part of the research design.

Part 1: audit the data before modeling

A good audit asks whether the dataset can support the intended prediction task.

Question	Why it matters
What does one row represent?	Defines the unit of observation and prevents mixed cases
Which column is the target?	Separates prediction objective from inputs
Which variables are numeric, categorical, datetime, text, or IDs?	Determines preprocessing and modeling choices
Are values missing?	May require imputation, removal, or redesign
Are categories consistent?	Avoids treating “A”, “a”, and “Batch A” as different categories
Are there outliers or impossible values?	May indicate errors, rare events, or valid extremes

A good data audit reduces the risk of training a high-scoring but meaningless model.

Part 2: clean missing values and outliers carefully

Missingness and extreme values can be errors, process signals, or meaningful rare events.

Missing values

Common actions:
- Remove only when justified
- Impute median or mode
- Add a missingness indicator
- Revisit data collection

Outliers

Possible actions:
- Correct impossible values
- Preserve rare valid cases
- Cap or transform values
- Use robust models

Key caution

Fit imputation and scaling rules only on training data, then apply them to validation/test data.

Example	Interpretation	Decision
Vibration missing during high-speed runs	Sensor saturation may be informative	Add missingness flag and investigate sensor
pH missing at random	Small random gaps	Median imputation may be acceptable
Final inspection score missing before inspection	Prediction-time leakage risk	Do not use as input

Part 3: choose transformations by data type

Models need numeric arrays, but disciplinary data often contain categories, IDs, dates, and text.

Data type	Examples	Typical handling
Numeric	temperature, pressure, speed	Impute missing values; scale if needed
Categorical	material type, machine ID, treatment group	One-hot encode or ordinal encode carefully
Datetime	timestamp, collection date	Extract elapsed time, hour, day, cycle, or lags
Identifier	part ID, student ID, batch ID	Usually remove; use for grouping or split strategy
Text	notes, descriptions, comments	Use separate NLP workflow; avoid accidental leakage

Question to ask: is this variable a meaningful predictor, an identifier, or a disguised answer?

Part 4: scale numeric features and encode categories

Transformations should make inputs usable without changing the meaning of the problem.

Scaling

Why it matters:
variables measured in large units can dominate distance-based or coefficient-based models.

Use for KNN, SVM, PCA, logistic regression, and regularized regression.

Encoding

Why it matters:
models cannot use raw text categories.

Use one-hot encoding for categories without natural order.
Be careful with ordinal encoding.

Pipeline rule

Fit scalers, encoders, and imputers on training data only.
Apply the learned transformations to validation and test sets.

Preprocessing rules are learned from the training set, not from the full dataset.

Part 5: data leakage can invalidate a model

Leakage creates inflated performance by letting the model see information unavailable in real use.

Target leakage

Input contains the answer or a near-answer.

Example: final inspection score used to predict pass/fail quality.

Time leakage

Future information is used to predict an earlier outcome.

Example: using post-failure data to predict pre-failure status.

Group leakage

Related samples appear in both train and test sets.

Example: parts from the same batch split across train and test.

Activity prompt

For each feature, ask: would this value be known at the moment the prediction is needed?

Part 6: engineer features with disciplinary meaning

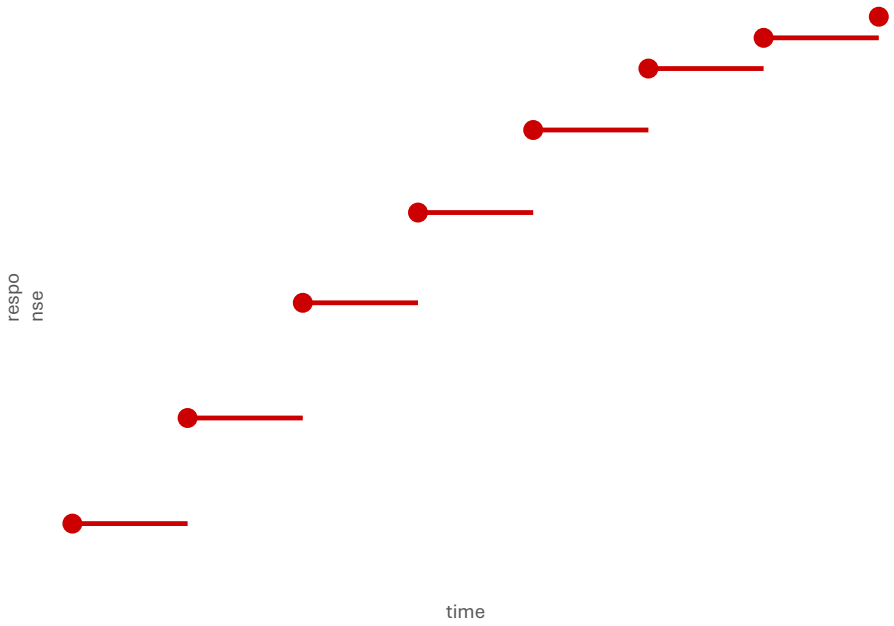
Feature engineering converts raw measurements into variables that better represent mechanisms, behavior, or context.

Raw data	Engineered feature	Why useful
Temperature time series	slope, max, time above threshold	Captures trend and thermal exposure
Sensor signal	rolling mean, variance, peak count	Summarizes stability and abnormal variation
Process settings	distance from nominal setting	Represents deviation from intended operation
Student activity logs	completion rate, delay, weekly activity	Summarizes engagement over time
Bioprocess trajectories	area under curve, growth rate, final-initial change	Captures dynamic process behavior

Good features make the model inputs closer to how a domain expert reasons about the system.

Mini-lesson: summarize time-series data into features

Many applied datasets start as trajectories but need one row per run, batch, part, or subject.



Initial / final

Starting value, ending value, final-initial change

Trend

Slope, growth rate, time to threshold

Stability

Mean, variance, coefficient of variation

Exposure

Area under curve, time above limit

Feature summaries should match the process behavior that matters for the outcome.

Part 7: select features to simplify the model

More variables are not always better. Selection can improve interpretation, cost, and robustness.

Filter methods

Use correlation, variance, or statistical association before training a model.

Embedded methods

Use model structure such as Lasso coefficients or tree-based importance.

Domain selection

Keep variables that are reliable, available early, and meaningful to collaborators.

Selection criteria

- Predictive value
- Interpretability
- Measurement cost
- Available at prediction time
- Robust across groups

Feature selection is a design decision, not only a statistical procedure.

Session 3 technical skill: pandas data audit

Goal: quickly inspect structure, missingness, variable types, and target balance.

```
import pandas as pd

df = pd.read_csv("process_data.csv")

print(df.shape)
display(df.head())

summary = pd.DataFrame({
    "dtype": df.dtypes.astype(str),
    "missing": df.isna().sum(),
    "unique": df.nunique()
})

display(summary)

# For classification targets
display(df["defect"].value_counts(normalize=True))
```

Use before modeling

This audit should happen before selecting an algorithm.

Instructor prompt

“What would stop us from trusting a model trained on this table?”

Notebook

Session 3 embedded notebook or data-preparation notebook

Session 3 technical skill: safe preprocessing pipeline

Goal: fit preprocessing decisions on training data only and apply them consistently to test data.

```
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder

numeric_features = ["temperature", "pressure", "vibration"]
categorical_features = ["material_type"]

preprocess = ColumnTransformer([
    ("num", Pipeline([
        ("impute", SimpleImputer(strategy="median")),
        ("scale", StandardScaler())
    ]), numeric_features),
    ("cat", OneHotEncoder(handle_unknown="ignore"),
     categorical_features)
])

# Later: Pipeline([("prep", preprocess), ("model", model)])
```

Why pipeline?

It prevents train/test contamination and documents the workflow.

Key principle

Never use test data to decide imputation, scaling, or encoding rules.

Link to baseline modeling

Use the preprocessing object inside the baseline model workflow.

Faculty activity: create a preprocessing plan

Use a workshop dataset or one from your own discipline.

Decision area	Participant response
Unit of observation	
Target variable	
Numeric inputs needing scaling	
Categorical inputs needing encoding	
Missing-data strategy	
Possible leakage variable	
Candidate domain-informed feature	
Feature to remove or postpone	

Deliverable: one preprocessing plan that could be implemented before model training.

Session 3 wrap-up: what you should take forward

Before modeling

- Define the unit of observation
- Identify target and inputs
- Audit data types
- Check missing values
- Check leakage risk

During preprocessing

- Fit transformations on training data
- Impute carefully
- Scale when needed
- Encode categories
- Preserve meaningful outliers

After preprocessing

- Document decisions
- Explain limitations
- Revisit data collection
- Engineer better features
- Prepare for model comparison

Next: Unsupervised Learning for Discovery — exploring structure when target labels are unavailable.